

Rajiv Gandhi University of Health Sciences, Karnataka

Fourth Semester B. Pharm Degree Examination – 06-Jun-2024

Time: Three Hours

Max. Marks: 75 Marks

PHYSICAL PHARMACEUTICS - II

Q.P. CODE: 5015

Your answers should be specific to the questions asked.

Draw neat labeled diagrams wherever necessary.

All the Questions are compulsory.

LONG ESSAYS

2 x 10 = 20 Marks

1. Define Colloids. Write its classifications and explain methods of preparation of colloids.

OR

Explain the accelerated stability testing for predicting shelf life of a drug. Give the limitations of the study.

2. Mention the equivalent diameters and explain the method of determination of particle size using coulter counter apparatus

SHORT ESSAYS

7 x 5 = 35 Marks

3. Explain optical properties of colloids.

OR

Differentiate colloidal dispersion and coarse dispersion.

4. Explain any one method for determination of viscosity of non-Newtonian fluids.

OR

What is Thixotrophy? Explain how it is measured.

5. Explain Heckel equation for different types of material.
6. Explain various frequency distribution curves for expressing particle size distribution.
7. Note on settling in suspension
8. Note on theories of emulsification
9. Discuss the protection of drugs against hydrolysis and oxidation.

SHORT ANSWERS

10 x 2 = 20 Marks

10. Define Rheopexy
11. Define bulk and true density
12. What is plug flow?
13. What do you mean by multi point viscometers? Give two examples.
14. Give two applications of micromeritics in pharmacy.
15. Define gold number.
16. Define Zero order reaction and give its equation.
17. Write Arrhenius equation and explain the terms.
18. Define flocculated suspension.
19. Give the fundamental properties of particles
